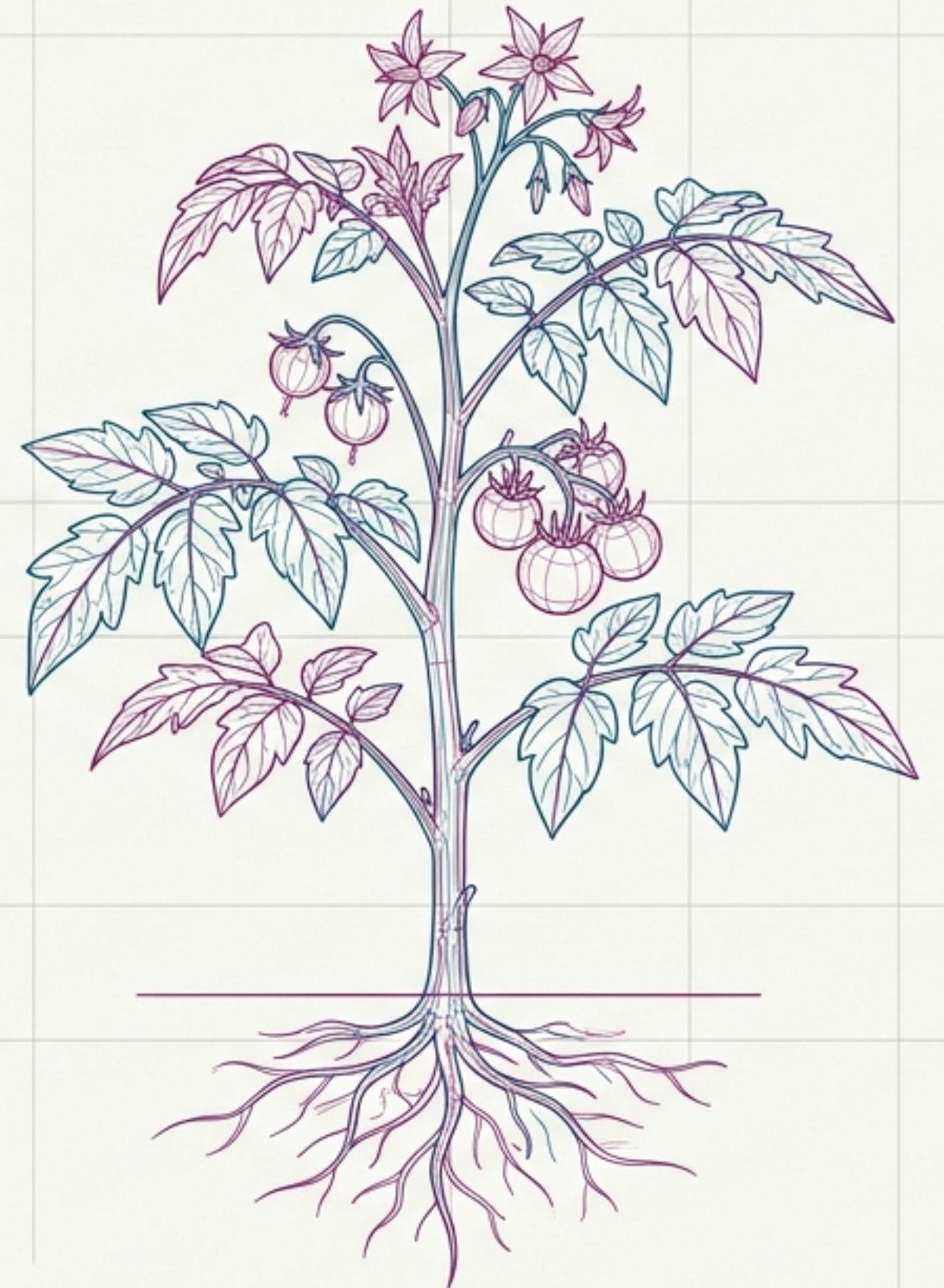


Integrated Multi-Omics of ISS VEG-05 Tomatoes

Decoding spaceflight dysbiosis and microbe-driven host defense.

Data sourced from NASA OSDR (OSD-766, OSD-767).



The spaceflight environment forces a deeply integrated biological reaction.



Microbiome Dysbiosis

The International Space Station environment fundamentally destabilizes the bacterial and fungal communities living on the plant.



Host Defense Activation

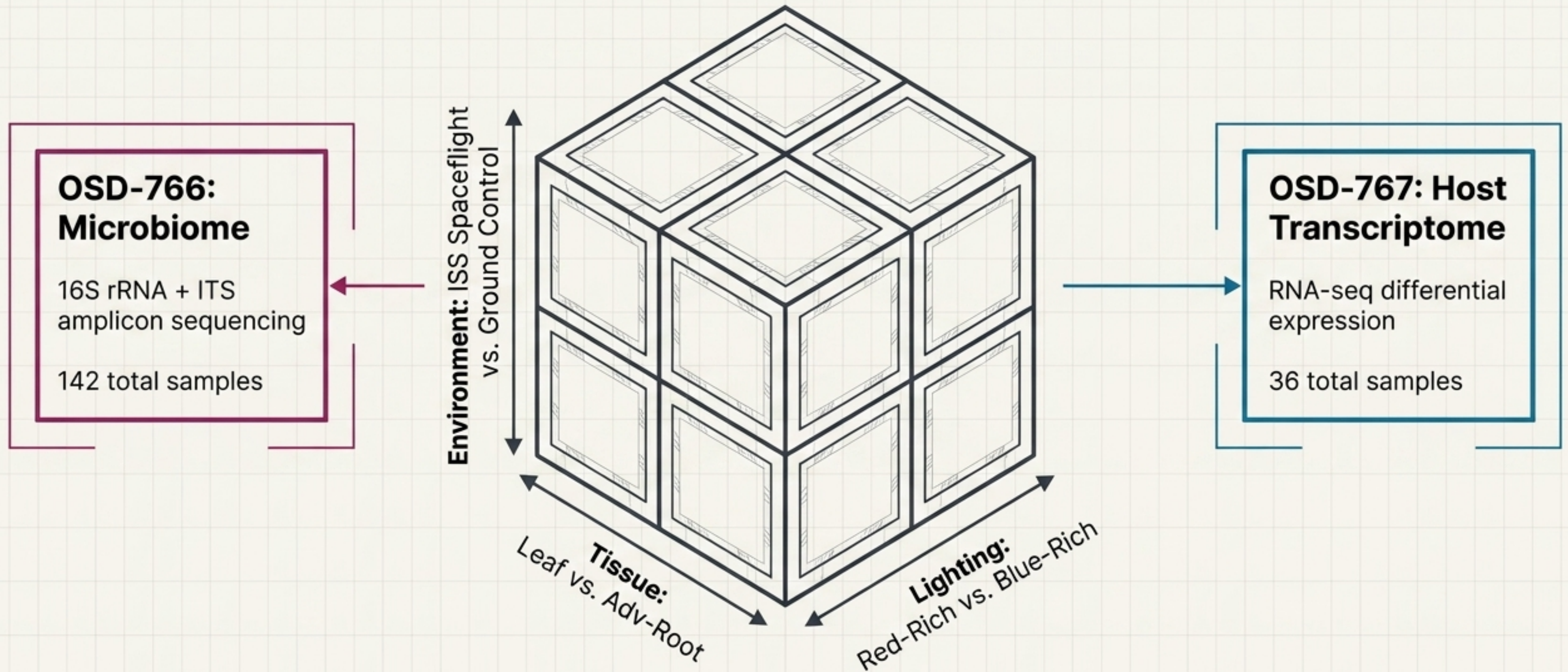
The plant undergoes massive genetic reprogramming, mounting a severe defense response, particularly in the root system.



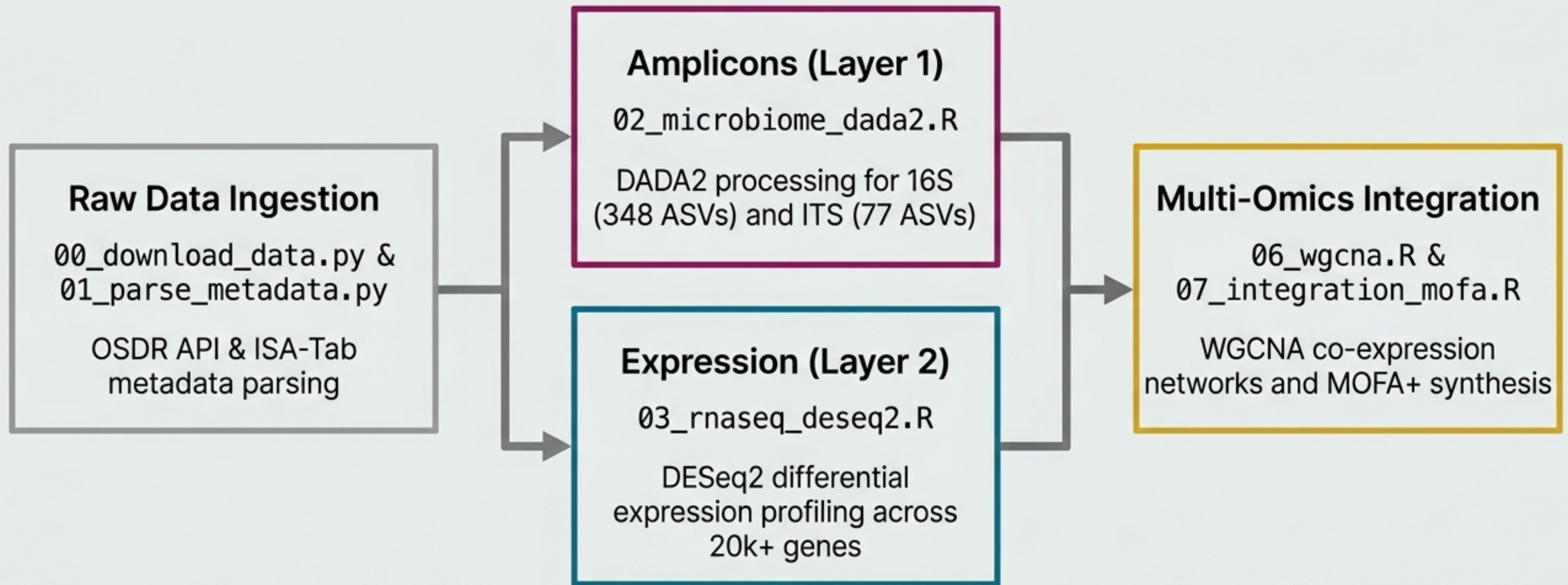
Microbe-Driven Mechanics

Multi-omics integration reveals the plant is actively fighting space-induced microbial dysbiosis, a response heavily magnified by blue lighting.

The VEG-05 Experimental Matrix maps environmental, lighting, and tissue variables.



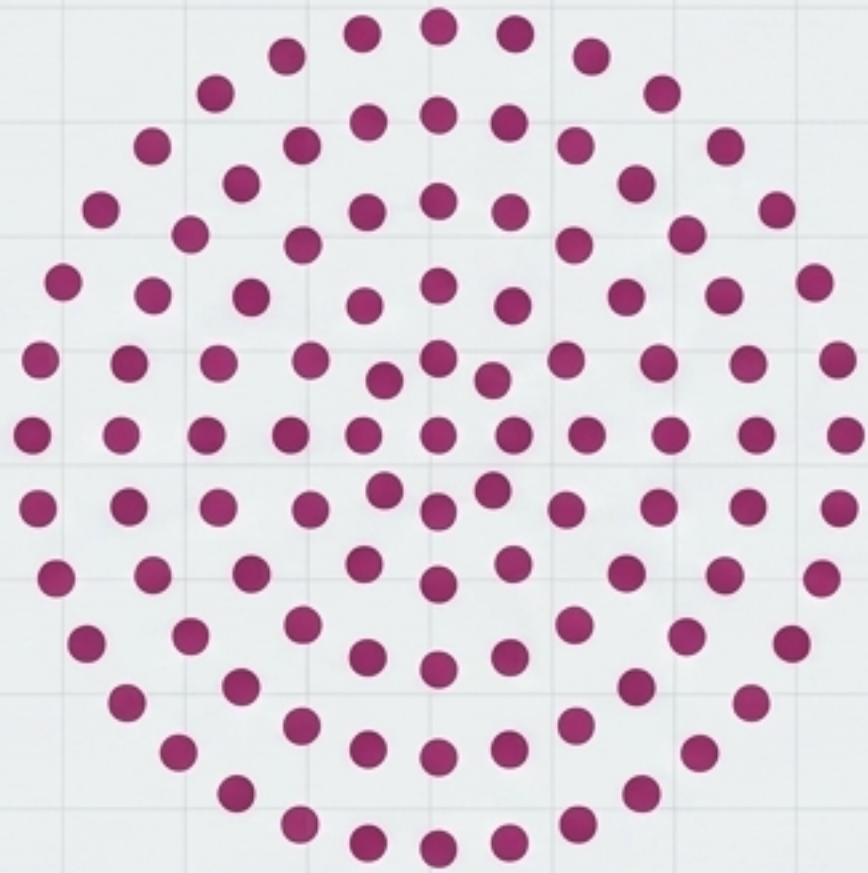
A reproducible computational engine drives raw OSDR data to biological synthesis.



Spaceflight significantly reshapes leaf bacterial and fungal communities

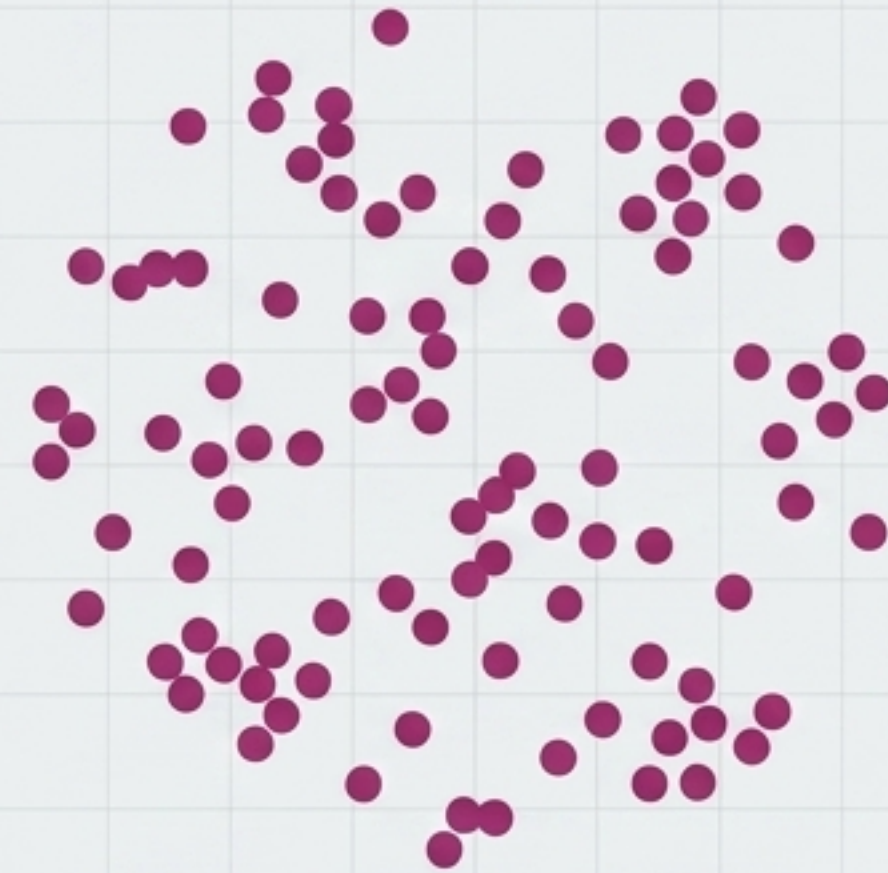
The ISS environment fundamentally destabilizes microbiome structure (PERMANOVA $p=0.001$, $R^2=0.42$).

Dysbiosis Index



Dysbiosis Score = 1.00 (± 0.47 for 16S, ± 0.24 for ITS)

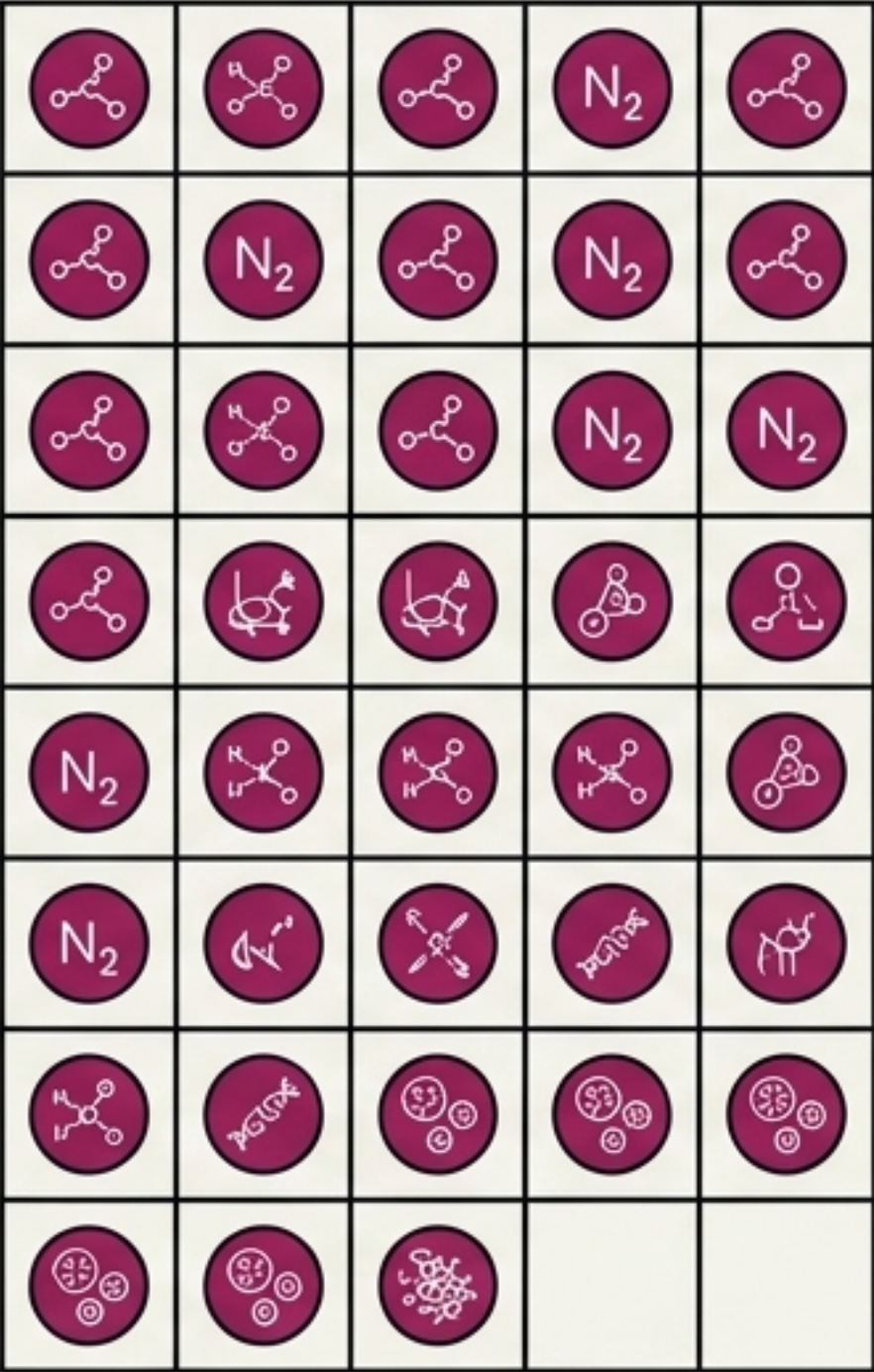
Ground Control Baseline



16S Dysbiosis Score = 2.06 (± 0.99)
ITS Dysbiosis Score = 3.64 (± 2.78)

ISS Flight Extremes

Aerobic chemoheterotrophs and potential plant pathogens dominate the flight microbiome.



FAPROTAX Predictions

29 functional profiles identified.
Bacterial shifts are heavily dominated by **aerobic chemoheterotrophy**, **nitrogen fixation**, and **methanotrophy**.

Dominant Bacterial Actors

Significant emergent representation of *Methylobacterium*, *Burkholderia*, and *Azospirillum* in flight samples.

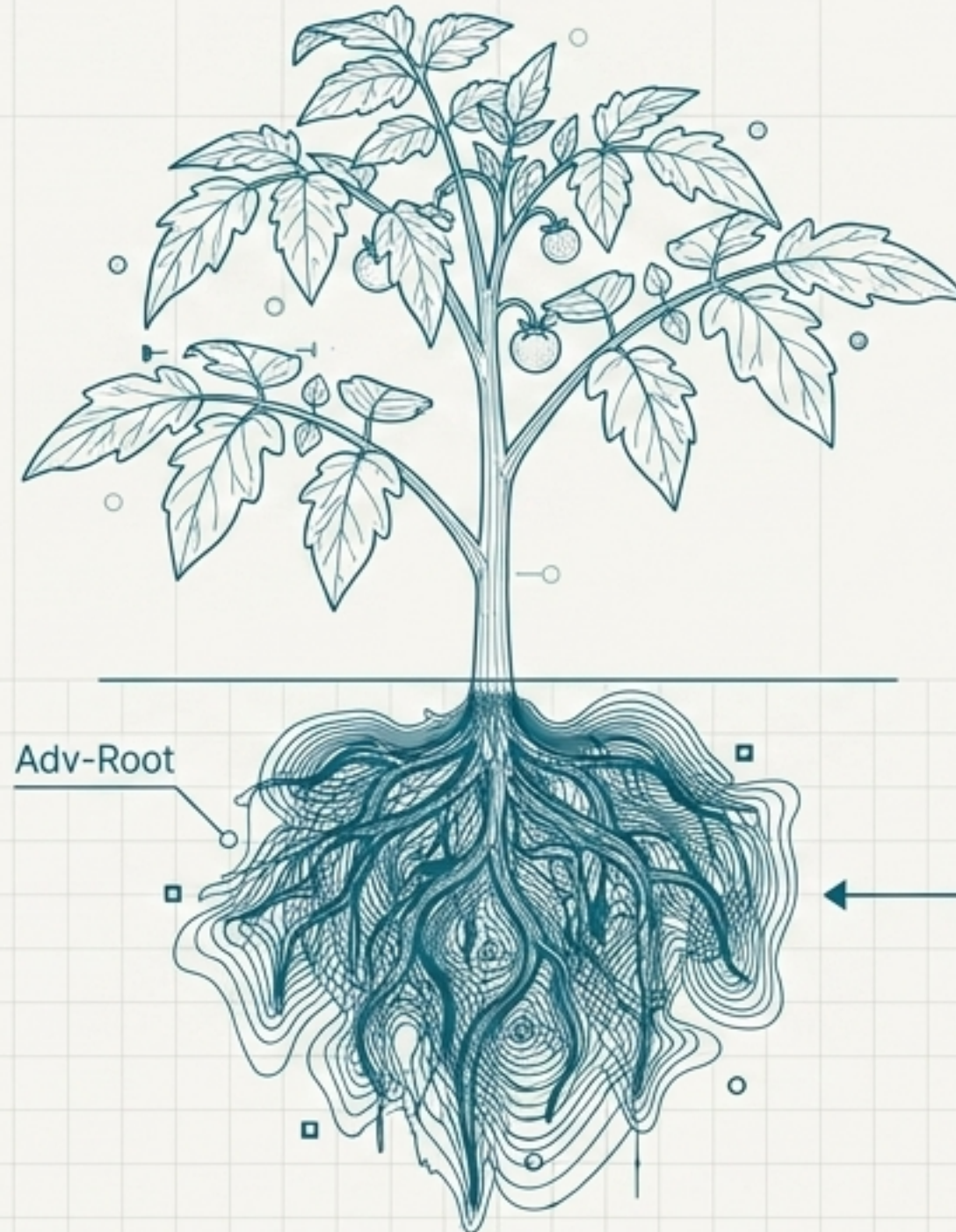
FUNGuild Pathogen Predictions

Identifies dominant **saprotrophs** (11 ASVs) and emergent **Plant Pathogens** (2 ASVs: *Fusarium* spp.) thriving in the space environment.

The host transcriptome reveals a massive, concentrated defense response in the roots.

Global Profiling Overview

Profiling 19,844 genes in the leaf tissue and 20,875 genes in the Adv-Root system.



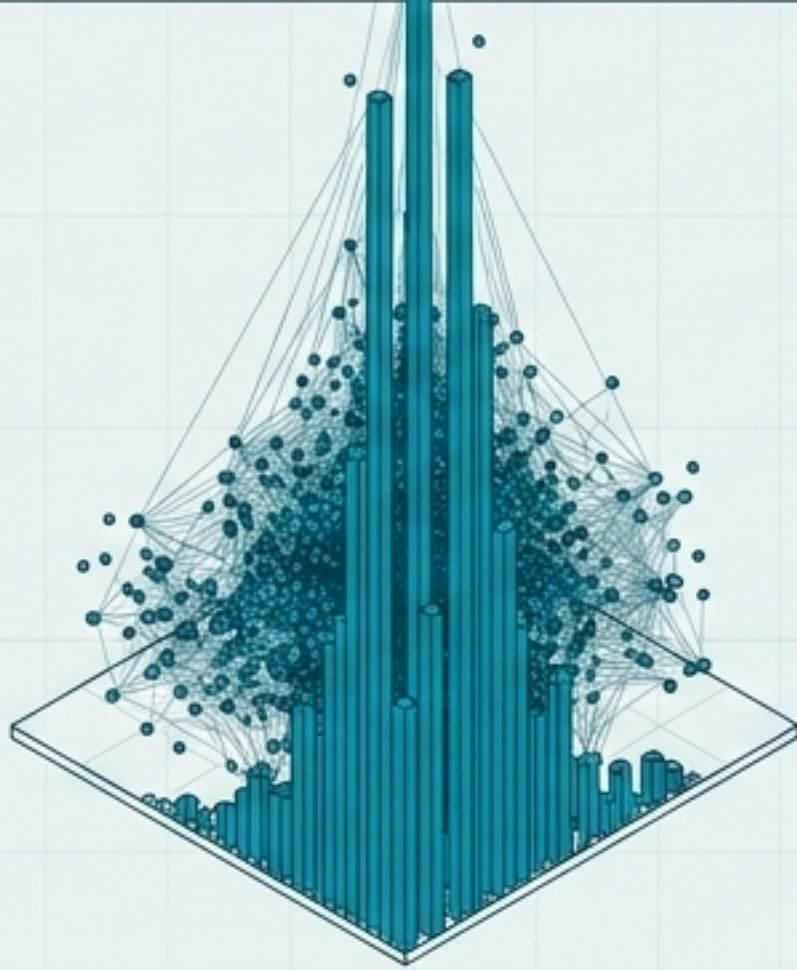
The Root Anomaly

The Adv-Root system exhibits a severe spaceflight effect with 896 Differentially Expressed Genes (DEGs).

Crucially, 87% of these 896 root DEGs are highly upregulated in flight, indicating an active, energy-intensive defense posture rather than general systemic failure.

Blue-rich lighting acts as a massive stress multiplier for flight leaves.

Flight × Light interaction in the leaf transcriptome reveals drastically different stress responses based on the lighting regime.



Blue-Rich Lighting Regime

4,716 DEGs

Triggered in spaceflight under blue-rich lighting.



Red-Rich Lighting Regime

523 DEGs

Triggered in spaceflight under red-rich lighting.

Insight: Blue light dramatically exacerbates the biological stress of the spaceflight environment compared to the red-light ground analog.

MOFA+ integration proves the host and microbiome are locked in a correlated response.

The Synthesis Model

Multi-Omics Factor Analysis (MOFA+) applied across 5 factors, 21 samples, and 3 views.

MOFA+
Layer Cake

The Synthesis Insight

This single integrated factor explains 48% of the total transcriptome variance, proving that the plant's gene expression is inextricably linked to the state of its microbiome.

Host Transcriptome

The Primary Driver

Factor 1 perfectly captures the spaceflight effect ($\rho = -0.76$, $p_{adj}=0.001$).

Microbiome

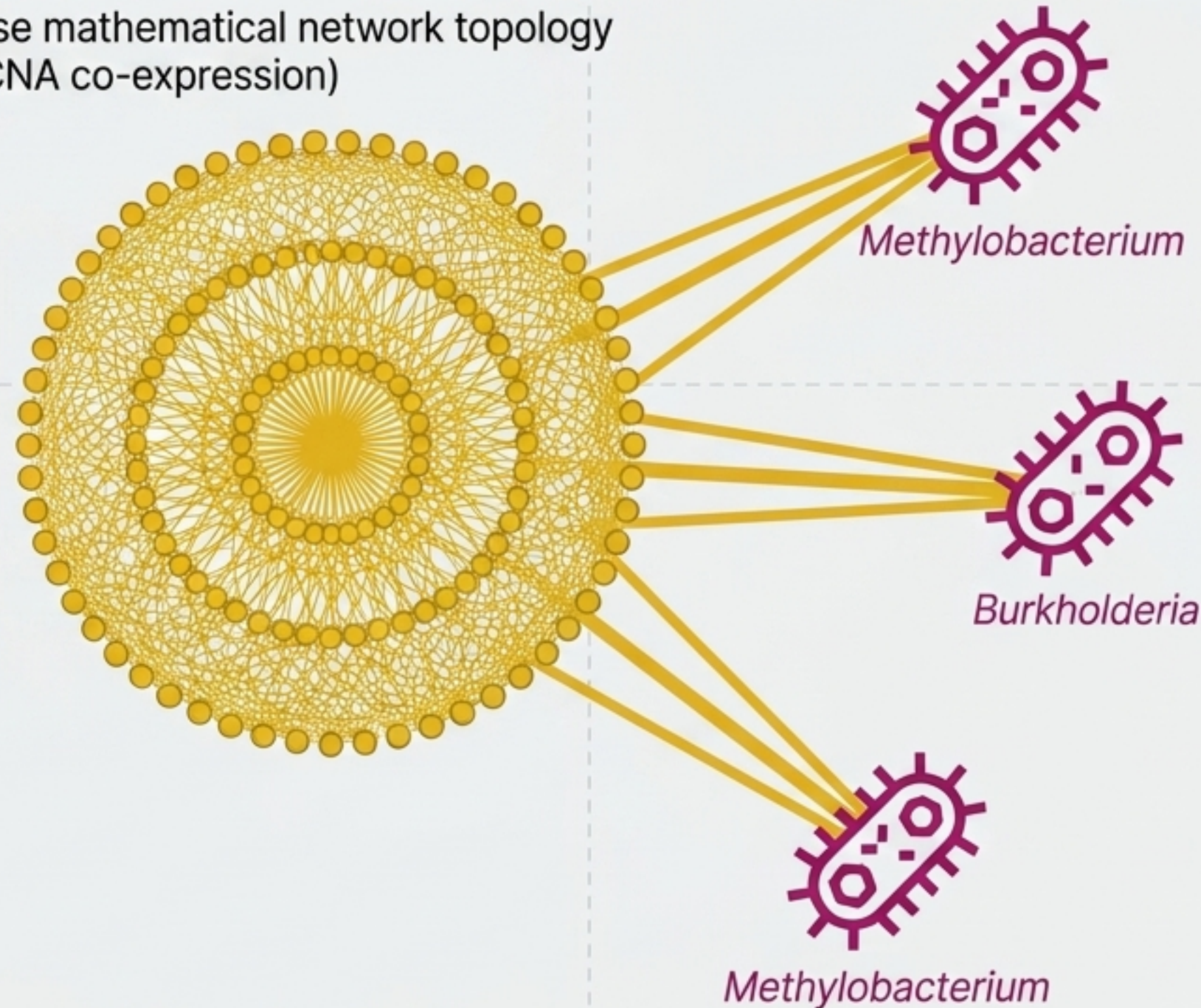
The Adv-Root “Black Module” isolates the precise microbe-driven defense network.

Network Statistics

WGCNA identified 13 leaf modules and 17 root modules. 29 significant module-taxon correlations were found (BH padj<0.05).

The Adv-Root Black Module isolates 169 specific genes.

Precise mathematical network topology (WGCNA co-expression)



The Culprit Mechanism

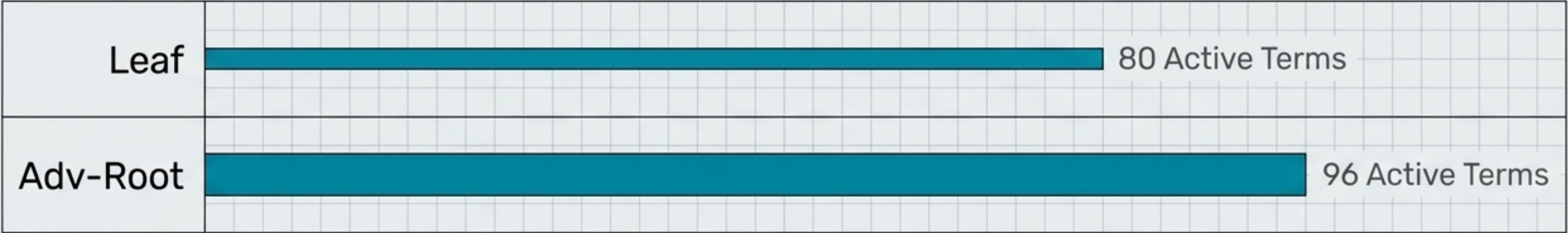
This specific 169-gene cluster has a near-perfect inverse correlation ($r = -0.85$) with ITS fungal dysbiosis.

It links specific taxa (*Methylobacterium*, *Burkholderia*) directly to the plant's defense response.

Crucially, this module does NOT correlate directly with microgravity itself—it is a dedicated biological reaction to microbial chaos.

Translating statistical networks into physical biological pathways.

Gene Ontology (GO) Enrichment reveals the specific functions of the activated genes.



Photosynthesis

Compensatory energy production rerouted to sustain plant defense.

Oxidative Stress

Active cellular defense mechanisms deployed against pathogen damage.

Ribosome Functions

Rapid protein synthesis for structural repair and enzyme production.

Insight: The integrated multi-omics data reveals a plant aggressively rerouting its energy from normal growth into combatting oxidative stress caused by erratic microbes.

Analytical constraints and dataset boundaries

To maintain rigorous scientific credibility, we transparently outline the limitations of the dataset.

Sample Size Constraints

Limited biological replicates (n=21 leaf, n=15 root for RNA-seq) constrain statistical power and broad generalizability.

Read Depth Reality

16S leaf samples yielded very low bacterial reads after host DNA removal (median 89 reads/sample).

Database Limits

FUNGuild utilized manual genus-level assignments; UNITE v7 was used due to local analysis environment access issues with v10.

Correlation vs. Causation

The integrated models are highly correlated mathematically, but physical experimental validation is required to prove direct causal relationships.

Fully reproducible open-source architecture.

```
// Open Source Pipeline Architecture
```

```
Custom Python FAPROTAX collapse scripts  
R integration pipelines (MOFA+, WGCNA)
```

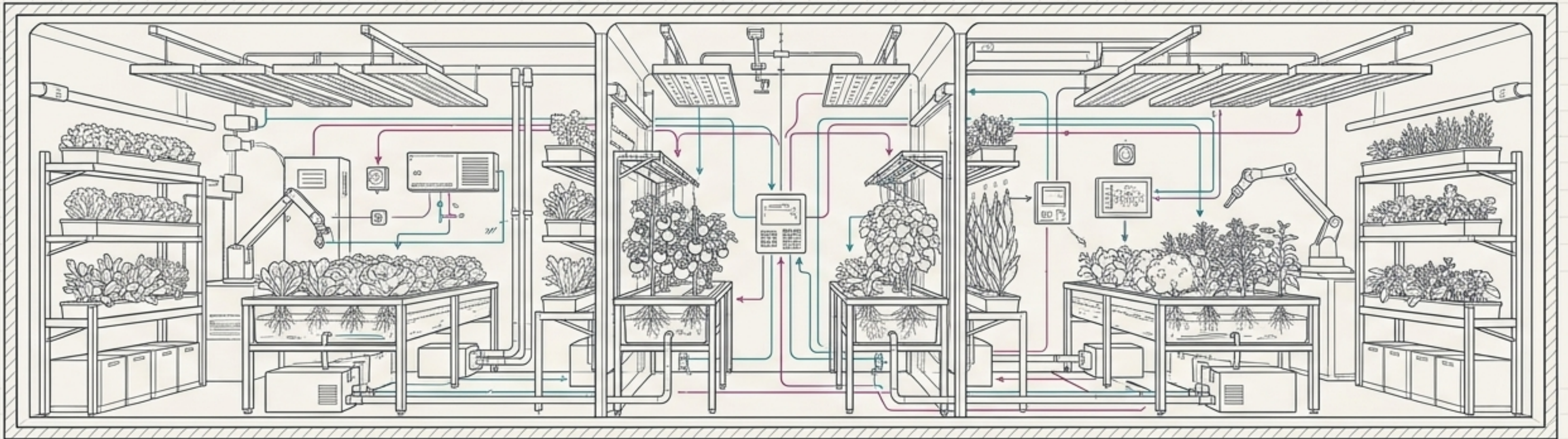
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Fully licensed under MIT
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Pre-wired DOI via Zenodo (ID 1287757020)  
NASA OSDR Citations (OSD-766, OSD-767)
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**Live Site: [dr-richard-barker.github.io/
veg05-integrated-omics](https://dr-richard-barker.github.io/veg05-integrated-omics)**

The future of space agriculture requires managing the invisible.



Sustaining crops on the International Space Station—and eventually Mars—cannot be achieved by managing the plant and the physical environment alone. We must actively monitor, model, and manage the highly volatile, integrated multi-omics bio-system.